Prayas JEE AIR (2025)

Weekly Test 29

Advanced

DURATION : 180 Minutes | **DATE : 11/01/2025** | **M. MARKS : 180**

ANSWER KEY

PHYSICS		CHEMISTRY		MATHEMATICS	
1.	(B)	18.	(A)	35.	(A)
2.	(A)	19.	(B)	36.	(A)
3.	(A)	20.	(D)	37.	(A)
4.	(A)	21.	(D)	38.	(A)
5.	(A, B)	22.	(B, C)	39.	(A, B, D)
6.	(A, B, C)	23.	(A, B, C)	40.	(A, D)
7.	(A, B)	24.	(A, C, D)	41.	(A, C, D)
8.	(7)	25.	(4)	42.	(5)
9.	(6)	26.	(160)	43.	(256)
10.	(6)	27.	(2)	44.	(2)
11.	(102)	28.	(25)	45.	(5)
12.	(4)	29.	(3)	46.	(5)
13.	(4)	30.	(9)	47.	(41)
14.	(2.00)	31.	(2.60)	48.	(97.00)
15.	(8.00)	32.	(42.50)	49.	(192.00)
16.	(3.00)	33.	(8.00)	50.	(105.00)
17.	(30.00)	34.	(7.10)	51.	(39.00)

PART-I (PHYSICS)

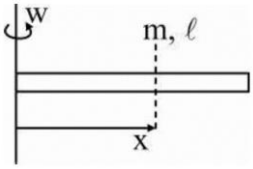
1. (B)

$$Q = \Delta U + W \Rightarrow C = C_V + \frac{RT}{V}\frac{dV}{dT}$$

$$\Rightarrow C = C_V - \frac{R}{\alpha V}(1-\alpha V)$$

2. (A)

Tension at point x = mω^2r



$$T = \left\{\frac{m}{\ell}(\ell - x)\right\}\cdot\omega^2\left\{\frac{\ell + x}{2}\right\}$$

$$T = \frac{m\omega^2}{2\ell}\left(\ell^2 - x^2\right)$$

3. (A)

$$x = A\sin\left(\frac{y}{B}\right)$$

$$\Rightarrow 1 = \frac{A}{B}\cos\left(\frac{y}{B}\right)\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \tan\theta = \frac{B}{A}\sec\left(\frac{y}{B}\right)$$

Thus, $\sin\theta = \dfrac{B\sec\left(\frac{y}{B}\right)}{\sqrt{A^2 + B^2\sec^2\left(\frac{y}{B}\right)}}$

Using Snell's law; $n(x)\sin\theta = n_0 \times \dfrac{B}{\sqrt{A^2 + B^2}}$

$$\Rightarrow n(x) = \frac{n_0}{\sin\theta}\times\frac{B}{\sqrt{A^2 + B^2}}$$

or $n(x) = n_0\left\{1 - \dfrac{x^2}{A^2 + B^2}\right\}^{1/2}$

4. (A)

$$E = \frac{Qx}{\frac{4}{3}\pi R^3 3\varepsilon_0} + \frac{Q/16}{4\pi\varepsilon_0 x^2}$$

$$\frac{dE}{dx} = 0$$

$$\Rightarrow \quad \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{R^3} - \frac{2}{16x^3}\right) = 0, \text{ at } x = \frac{R}{2}$$

E is minimum.

5. (A, B)

$$dA = 2\pi\times dy\times\frac{1}{\cos\theta} \text{ and}$$

$$d\tau = \eta\frac{2\pi\times\sqrt{R^2 + h^2}}{h}\times\frac{V}{t}dy:$$

$$d\tau = \frac{\eta 2\pi}{ht}\times\sqrt{R^2 + h^2}\times\omega\cdot x^3 dy$$

$$\because \quad \tan\theta = \frac{R}{h} = \frac{x}{y} \quad ; dy = \frac{h}{R}dx$$

$$\Rightarrow \tau\frac{2\pi\eta\omega\sqrt{R^2 + h^2}}{Rt}\int_0^R x^3 dy, P = \tau\omega$$

6. (A, B, C)

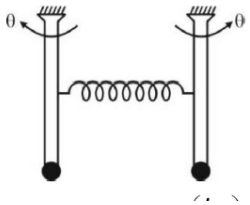
W_L: The rods oscillate independently

$$\Rightarrow \quad T = 2\pi\sqrt{\frac{l}{Mgd}}$$

$$\Rightarrow \quad w = \sqrt{\frac{Mgd}{l}}$$

$$= \sqrt{\frac{2M.g.\frac{3L}{4}}{ML^2 + \frac{ML^2}{3}}} = \sqrt{\frac{9}{8}\frac{g}{L}}$$

W_H:



Tension in the spring = $2.\left(\dfrac{L}{2}\theta\right) = L\theta$

$\Rightarrow$ Restoring torque on one rod

$$= k\left(L\theta\right).\frac{L}{2} + Mg.\frac{L}{2}\theta + MgL\theta$$

$$= \left[\frac{kL^2}{2} + \frac{MgL}{2} + MgL\right]\theta$$

$$\Rightarrow \quad -\left[\frac{kL^2}{2} + \frac{3MgL}{2}\right]\theta = l\alpha$$

$$\Rightarrow \quad \left[\frac{ML^2}{3} + ML^2\right]\alpha = -\left[kL + 3Mg\right].\frac{L}{2}\theta$$

$$\Rightarrow \quad \alpha = -\frac{3}{8}\left[\frac{kL + 3Mg}{ML}\right]\theta$$

$$\Rightarrow \quad \alpha = -\frac{3}{8}\left[\frac{k}{M} + \frac{3g}{L}\right]\theta$$

$$\Rightarrow \quad W_H = \sqrt{\frac{3}{8}\left[\frac{k}{M} + \frac{3g}{L}\right]}$$

7. (A, B)

$$|F| = -\frac{\partial U}{\partial r} = \frac{Ke^2}{r^4}$$

$$F = \frac{mv^2}{r}$$

$$\frac{Ke^2}{r^4} = \frac{mv^2}{r} \qquad ...(1)$$

$$mvr = \frac{nh}{2\pi} \qquad ...(2)$$

From (1) and (2)

$$\frac{Ke^2}{\left(\frac{nh}{2\pi mv}\right)^3} = mv^2$$

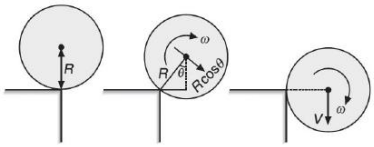
$$\frac{8\pi^3\ m^3v^3Ke}{n^3\ h^3} = mv^2$$

$$v = \frac{n^3\ h^3}{8\pi^3\ m^2Ke^2}$$

$$E \propto mv^2 \propto \frac{n^6}{m^3}$$

8. (7)

The cylinder rotates about the point of contact as shown in Figure.



Hence, the mechanical energy of the cylinder will be conserved i.e.,

$(PE + KE) = (PE + KE)_2$

$$\Rightarrow \quad mgR + 0 = mgR\cos\theta + \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2$$

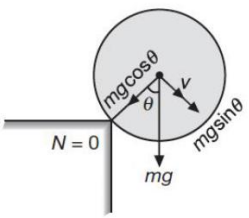
where, $I = \frac{1}{2}mR^2$ and for no slipping at point of contact, we have $\omega = \frac{v}{R}$

$$\Rightarrow \quad mgR = mgR\cos\theta + \frac{1}{2}\left(\frac{1}{2}mR^2\right)\left(\frac{v^2}{R^2}\right) + \frac{1}{2}mv^2$$

$$\Rightarrow \quad \frac{3}{4}v^2 = gR(1-\cos\theta)$$

$$\Rightarrow \quad v^2 = \frac{4}{3}g\mathrm{R}(1-\cos\theta)$$

$$\Rightarrow \quad \frac{v^2}{R} = \frac{1}{4}g(1-\cos\theta) \qquad \ldots\ldots(1)$$



At the time of leaving contact, normal reaction N = 0 and $\theta = \theta_c$ hence,

$$mg\cos\theta = \frac{mv^2}{R}$$

$$\Rightarrow \quad \frac{v^2}{R} = g\cos\theta \qquad \ldots\ldots(2)$$

From equation (1) and (2),

$$\frac{4}{3}g(1-\cos\theta_c) = g\cos\theta_c$$

$$\Rightarrow \quad \frac{7}{4}\cos\theta_c = 1$$

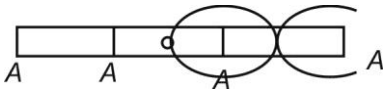
$$\Rightarrow \quad \cos\theta_c = \frac{4}{7}$$

9. (6)

Velocity of the longitudinal waves in the rod = $v = \sqrt{Y/d} = \sqrt{2\times10^{11}/8000} = 5000$ m/s .

The wavelength of the wave for the mode of vibration in which 2 antinodes occur, is

$$\lambda = \frac{1.25}{(3/2)} = \frac{5}{6}\ \mathrm{m}$$

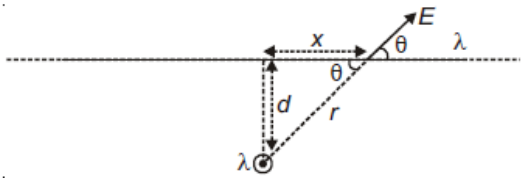


Hence, frequency of vibration

$$n = \frac{V}{\lambda} = \frac{5000}{5/6}\ \mathrm{Hz}$$

= 6000 Hz

10. (6)



$$E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

$\therefore\ dF = E\ \mathrm{sinq}\ (\mathrm{l}\ dx)$

$$= \frac{\lambda}{2\pi\varepsilon_0 r}\times\left(\frac{d}{r}\right)\times\lambda dx$$

$$F = \frac{\lambda^2 d}{2\pi\varepsilon_0}\int_{-\infty}^{\infty}\frac{dx}{d^2+x^2} = \frac{\lambda^2}{2\varepsilon_0}$$

11. (102)

The β^+ decay equation is

$${}^{64}_{29}Cu \rightarrow {}^{64}_{28}Ni + {}^{0}_{1}e + v + Q_1$$

The energy of the reaction is

$$Q_1 = (m_{cu} - m_{Ni} - 2m_e)c^2$$

Where 'm' represents the atomic mass. Emitted neutrino can have KE

$$0 \le k_1 < Q_1$$

The EC decay equation is

$^{64}_{29}Cu + ^{0}_{-1}e \rightarrow ^{64}_{28}Cu + \nu + Q_2$

The energy of the reaction is

$Q_2 = (m_{Cu} - m_{Ni})c^2$

The neutrino can have KE

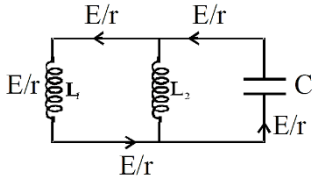
$0 \le k < Q_2$

$\therefore\ k_2 - k_1 = Q_2 - Q_1 = 2m,c^2 = 1.02\,MeV$

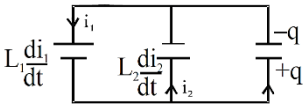
k = 1.02 and 100k = 102

12. (4)

$t = 0$

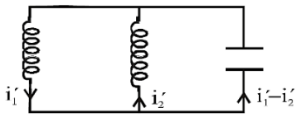


general $t = t$



So, $L_1\dfrac{di_1}{dt} = -L_2\dfrac{di_2}{dt} = \dfrac{-q}{C}$

for maximum current in L_2.



$\Rightarrow L_1\left(i_1' - \dfrac{E}{r}\right) = -L_2\left(i_2' - 0\right)$ …(i)

$\dfrac{1}{2}L_1\left(\dfrac{E}{r}\right)^2 = \dfrac{1}{2}L_1 i_1'^2 + \dfrac{1}{2}L_2 i_2'^2$ …(ii)

$\Rightarrow \dfrac{L_1E^2}{r^2} = L_2 i_2'^2 + L_1\left(\dfrac{E}{r} - \dfrac{L_2 i_2'^2}{L_1}\right)^2$

$\Rightarrow i_2' = \dfrac{2EL_1}{r(L_1 + L_2)}$

E = 5V

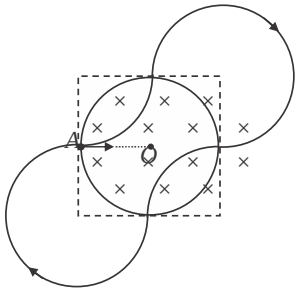
L_1 = 4 mH

L_2 = 1 mH

$r = 2\Omega$

$\Rightarrow i_2' = \dfrac{2\times5\times4}{5\times2} = 4\,\text{A}$

13. (4)

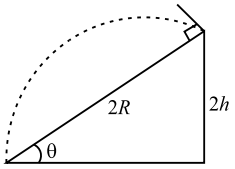


Path of the particle is given below

Time to complete this path

$T = \dfrac{4\pi m}{qB} = 4\pi\,\text{sec.}$

14. (2.00)



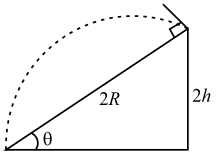
$\dfrac{2\omega R}{g\cos\theta}\times\omega R\sin\theta = 2R\cos\theta$ … (i)

$mg\sin\theta = m\omega^2 R$ …. (ii)

$\Rightarrow \tan\theta = 1$ or $\sin\theta = \dfrac{1}{\sqrt{2}}$

$\Rightarrow \omega = \sqrt{\dfrac{g}{\sqrt{2}R}}$ and $h = \dfrac{R}{\sqrt{2}}$

15. (8.00)



$\dfrac{2\omega R}{g\cos\theta}\times\omega R\sin\theta = 2R\cos\theta$ … (i)

$mg\sin\theta = m\omega^2 R$ …. (ii)

$\Rightarrow \tan\theta = 1$ or $\sin\theta = \dfrac{1}{\sqrt{2}}$

$\Rightarrow \omega = \sqrt{\dfrac{g}{\sqrt{2}R}}$ and $h = \dfrac{R}{\sqrt{2}}$

16. (3)

$\Rightarrow (\mu - 1)\dfrac{5}{2} = d\sin\theta$

$\Rightarrow \left(\dfrac{3}{5}\right)\left(\dfrac{5}{2}\right) = 3\sin\theta$

$\Rightarrow \sin\theta = \dfrac{1}{2}$

$\theta = 30°$ (not small angle)

$\Rightarrow \tan\theta = \dfrac{y}{b}.$

$\Rightarrow y = \dfrac{1}{\sqrt{3}}\,m$

x = 3

17. (30)

$(\mu - 1)\left(\dfrac{5m}{2}\right) = n\lambda$

$\Rightarrow \dfrac{3}{2}\times10^{-3} = n\left(500\times10^{-9}\right)$

$\Rightarrow n = 3000$

$100\,y = 3000$

y = 30

PART-II (CHEMISTRY)

18. (A)

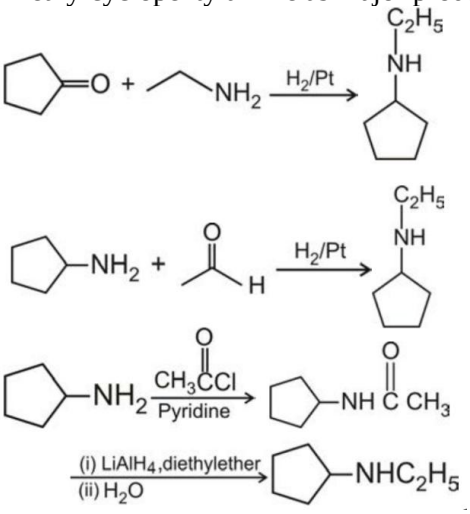
Λ_m of $HCl > KCl > LiCl > CH_3COOH$

19. (B)

Kuchervo reaction

20. (D)

Reaction (a, b, c) all give
N-ethylcyclopentylamine as major product.



21. (D)

$$P.E. = \frac{kq_1q_2}{r} = -\frac{1}{4\pi\varepsilon_0} \times \frac{3e \times e}{a_0 \times \frac{(2)^2}{3}}$$

22. (B, C)

$$P_{total} = \frac{P_A}{Y_A} = \frac{P_B}{Y_B}$$

$$\frac{X_A P_A^\circ}{Y_A} = \frac{(1-X_A)P_B^\circ}{(1-Y_A)}$$

$$\frac{1}{X_A} = \frac{1}{Y_A} \times \frac{P_A^\circ}{P_B^\circ} + \frac{P_B^\circ - P_A^\circ}{P_B^\circ}$$

$$= \frac{P_A^\circ}{P_B^\circ}, \text{ Intercept} = \frac{P_B^\circ - P_A^\circ}{P_B^\circ}$$

23. (A, B, C)

$Cu^{+2} = [Ar]\ 3d^9$ — $\uparrow\downarrow$ 1 ; $\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$

$Mn^{+3} = [Ar]\ 3d^4$ — 1 $-$; 1 1 1

24. (A, C, D)

As the positive charge on species increases, ionisation energy increases.

25. (4)

1, 3, 5, 7 statement are correct.

26. (160)

$$C(s) + O_2(g) \rightarrow CO(g) + CO_2(g)$$

a mole b mole

POAC on carbon, $1 \times 0.1 = a + b$

POAC on oxygen, $2 \times 0.08 = a + 2b$

$a = 0.04 \& b = 0.06$

$$5CO(g) + I_2O_5(s) \rightarrow 5CO_2(g) + I_2$$

$0.04/5$ mole

gmequivalent of I_2 = gmequivalent of hypo

$0.008 \times 2 = 0.1 \times x \times 10^{-3}$

x = 160 ml

27. (2)

M. wt 86 of ketone

So m. formula = $C_5H_{10}O$

(1) $Et-\underset{\underset{O}{\|}}{C}-Et \xrightarrow{NaBH_4} Et-\underset{\underset{OH}{|}}{CH}-Et$

Optically inactive compound

(2) $CH_3-CH_2-CH_2-\underset{\underset{O}{\|}}{C}-Me \xrightarrow{NaBH_4}$

$CH_3-CH_2-CH_2-\underset{\underset{OH}{|}}{CH}-Me$ (±) racemic mixture

(3) $CH_3-\underset{\underset{H_3C}{|}}{CH}-\underset{\underset{O}{\|}}{C}-Me \xrightarrow{NaBH_4}$

$CH_3-\underset{\underset{H_3C}{|}}{CH}-\underset{\underset{OH}{|}}{CH}-Me$ (±) racemic mixture

(2 ; 3)

28. (25)

$$Z = \frac{PV}{nRT}$$

$$\frac{V}{n} = \frac{ZRT}{P} = \frac{0.8 \times 0.0821 \times 300}{8}$$

= 2.463 litre/mole

$\therefore$ Volume of 10 mol of the gas = 24.63 litre.

29. (3)

NH_4NO_2, $(NH_4)_2Cr_2O_7$, NH_4MnO_4

30. (9)

x = 2, y = 3, z = 4

31. (2.60)

Rate law = $k^1 [CHCl_3][Cl_2]^{1/2}$

$$k^1 = k_2\left(\frac{k_1}{k_{-1}}\right)^{1/2} = 1.3 \times 10^{-2} \times \left(\frac{4.8 \times 10^3}{1.2 \times 10^3}\right)^{1/2}$$

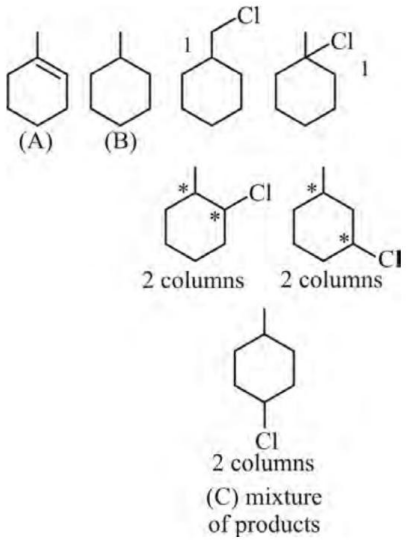
$= 2.6 \times 10^{-2}\ M^{-1/2}\ sec^{-1}$.

32. (42.50)

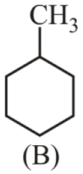
$E_{a\,overall} = E_{a2} + \frac{1}{2}(E_{a1} - E_{a-1})$

= 45 + 0.5 (10 – 15) = 42.5 kJ/mol.

33. (8.00)



34. (7.10)



M = 98 gm $\quad n_B = \frac{9.8}{98} = 0.1$

$n_B = n_{Cl_2} = 0.1$

$W_{Cl_2} = 0.1 \times 71 = 7.10\,gm$

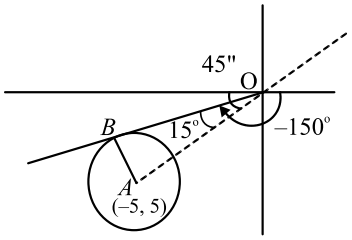
PART-III (MATHEMATICS)

35. (A)

$|z + 5 + 5i| \le \frac{5\sqrt{3} - 1}{2}$

Point *B* has least principal argument *AB*

$= \frac{5}{2}(\sqrt{3} - 1)$



$OA = 5\sqrt{2}$

$\angle AOB = \frac{\pi}{12}$

$\therefore \arg(z) = \frac{-5\pi}{6}$

36. (A)

Put $x = \frac{1}{y}$

$$\Rightarrow \lim_{x\to\infty}\left(\frac{x^2\sin\left(\frac{1}{x}\right) - x}{|x|}\right) = \lim_{y\to 0}\left(\frac{\sin y - y}{y}\right) = 0$$

37. (A)

Slope of tangent = 1

Equation of tangent to parabola at (*a*, 2*a*) is $y - 2a = x - a\,(x + a)$

$\Rightarrow y - x - a = 0$

Equation of circle touching parabola at (*a*, 2*a*) is

$(x - a)^2 + (y - 2a)^2 + \lambda\,(y - x - a) = 0$

It passes through (0, 0)

$a^2 + 4a^2 + \lambda\,(-a) = 0 \Rightarrow \lambda = 5a$

So, equation of circle is $x^2 + y^2 - 7ax - ay = 0$

On comparing with general equation of circles

$$\text{Radius of circle} = \sqrt{\frac{49a^2}{4} + \frac{a^2}{4}} = \frac{5a}{\sqrt{2}}$$

38. (A)

$AB = A + B$

$\Rightarrow B = AB - A = A(B - I)$

$\Rightarrow \det(B) = \det(A)\,\det(B - I) = 0$

$$\left[\because A^{2006} = 0 \Rightarrow \det\left(A^{2006}\right) = 0 \Rightarrow \det(A) = 0\right]$$

$\Rightarrow \det(B) = 0$

39. (A, B, D)

Let $\frac{dy}{dx} + y = f(x)$ is linear differential equation

I.F. = e^x

$\therefore$ Solution is $ye^x = \int e^x f(x)dx + C$

Now if $0 \le x \le 2$ then $ye^x = \int e^x e^{-x}dx + C$

$\Rightarrow ye^x = x + C$

$y(0) = 1, \Rightarrow C = 1$

$\therefore ye^x = x + 1$

$\therefore\ y = \dfrac{x+1}{e^x};$

$\therefore\ y(1) = \dfrac{2}{e}$

Also $y' = \dfrac{e-2e}{e^2} = \dfrac{-e}{e^2} = -\dfrac{1}{e}$

If $x > 2$

$ye^x = \int e^{x-2}dx$

$\therefore\ ye^x = e^{x-2} + C$

$\therefore\ y = e^{-2} + Ce^{-x}$

As y is continuous

$\therefore\ \lim\limits_{x\to 2} \dfrac{x+1}{e^x} = \lim\limits_{x\to 2}(e^{-2} + Ce^{-x})$

$\therefore\ 3e^{-2} = e^{-2} + Ce^{-2} \Rightarrow C = 2$

$\therefore$ for $x > 2$

$y = e^{-2} + 2e^{-x}$

$\Rightarrow y(3) = 2e^{-3} + e^{-2} = e^{-2}(2e^{-1} + 1)$

$\Rightarrow\ y' = -2e^{-x}$

$\Rightarrow\ y'(3) = -2e^{-3}$

40. (A, D)

(A) For $I_2 = \int\limits_0^2 \sin x^2 dx$; Put $x = t + 1$

$= \int\limits_0^1 \sin(x+1)^2 dx \Rightarrow I_1 - I_2 = 0$

(D) $4(2x-1) = k \Rightarrow dx = \dfrac{1}{8}dk$

$\therefore I_2 = 8\int\limits_0^1 \cos[16(2x-1)^2]dx$

$= \int\limits_{-4}^4 \cos k^2 dk = I_1$

$\therefore I_1 - I_2 = 0$

Note: in each option, L.H.S is taken as I_1 and R.H.S as I_2

41. (A, C, D)

$g(x + y) = e^x g(y) + e^y g(x),\ g(0) = 0$

$g'(x) = \lim\limits_{h\to 0} \dfrac{g(x+h) - g(x)}{h}$

$= \lim\limits_{h\to 0} \dfrac{e^x g(h) + e^h g(x) - g(x)}{h}$

$g'(x) = 2e^x + g(x)$

$\Rightarrow \dfrac{dy}{dx} - y = 2e^x$

$\Rightarrow e^{-x}\dfrac{dy}{dx} - e^{-x}y = 2 \Rightarrow \dfrac{d}{dx}\left(e^{-x}y\right) = 2$

$\Rightarrow e^{-x}y = 2x + c \Rightarrow g(x) = 2xe^x$

$\lim\limits_{x\to 0} \dfrac{g(x)}{x} = 2, g'(x) = 2e^x + 2xe^x = 2e^x(x+1)$

$\Rightarrow$ g $\uparrow$ for $x \in (-1, \infty)$

Rane = $\left[\dfrac{-2}{e}, \infty\right)$

42. (5)

By observation, $x = -1$ is one root of equation

$x^3 + 3x^2 + 5x + 3 = 0$

$\therefore x^3 + 3x^2 + 5x = 3 = 0$

$\Rightarrow (x + 1)(x^2 + 2x + 3) = 0$

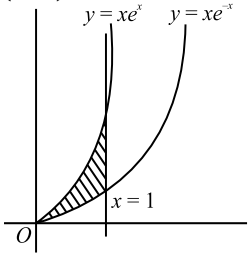
Now, roots of equation $x^2 + 2x + 3 = 0$ are nonreal

So, the equation $x^2 + px + q = 0$ has both roots common with $x^2 + 2x + 3 = 0$

Hence, $\dfrac{1}{1} = \dfrac{p}{2} = \dfrac{q}{3} \Rightarrow p = 2,\ q = 3$

So, $(p + q) = 5$

43. (256)



Required area $\lambda = \int\limits_0^1 \left(xe^x - xe^{-x}\right)dx$

$= \left\{x\left(e^x + e^{-x}\right) - \left(e^x - e^{-x}\right)\right\}_0^1$

$= \left\{\left(e + e^{-1}\right) - \left(e - e^{-1}\right)\right\} - \left\{0 - (1-1)\right\}$

$= \dfrac{2}{e}$

$\therefore\quad 128e\lambda = 128e \times \dfrac{2}{e} = 256$

44. (2)

Let r be radius and h be height of the cylinder

$\therefore volume(v) = \dfrac{1}{2}\pi r^2 h(given)$

Total surface area

$S = \dfrac{1}{2}(2\pi rh) + 2\left(\dfrac{1}{2}\pi r^2\right) + 2rh$

$= (\pi + 2) + \pi r^2$

$\therefore \dfrac{ds}{dr} = \dfrac{-2v(\pi+2)}{\pi r^2} + 2\pi r$ ....(1)

$\therefore \dfrac{d^2s}{dr^2} = \dfrac{4v(\pi+2)}{\pi r^3} + 2\pi$ .....(2)

For maximum or minimum

$\dfrac{ds}{dr} = 0 \Rightarrow \pi r^3 = \dfrac{v(\pi+2)}{\pi}$

Hence S is minimum when

$-\pi r^2 h(\pi + 2) + 2\pi^2 r^3 = 0$

$\Rightarrow \dfrac{h}{2r} = \dfrac{\pi}{\pi + 2} \qquad \therefore k = 2$

45. **(5)**

$4+\lambda_1+1+2\lambda_2 \Rightarrow \lambda_1-2\lambda_2=-3$ .....(1)

$-3-4+\lambda_1=-1-3\lambda_2 \Rightarrow 4\lambda_1-3\lambda_2=-2$ .....(1)

From (1) and (2),

$-5\times 2=-10$

$\lambda_2=2, \lambda_1=1$

$\Rightarrow$ point of intersection (4 + 1, – 3 – 4, – 1 +7)

i.e. (5, – 7, 6).

Distance of (5, –7, 6) from (1, –4, 7)

$=\sqrt{16+9+1}=\sqrt{26}$

46. **(5)**

$$4\left[\frac{\sin x+\sqrt{3}\cos x}{2}\right]\left[\frac{\sqrt{3}\cos x+\sin x}{2}\right]$$

$=a^2+\sqrt{3}\sin 2x-\cos 2x$

$2\cos 2x + 2 = a^2$

$$-1\le\frac{a^2-2}{2}\le 1$$

$0 \le a^2 \le 4$

$–2 \le a \le 2$

47. **(41)**

$$\alpha=\lim_{r\to\infty}\sum_{n=1}^{r}\tan^{-1}\left\{\frac{16(2n+1)}{(2n+1)^4-4(2n+1)^2+16}\right\}$$

$$=\lim_{r\to\infty}\sum_{n=1}^{r}\tan^{-1}\left\{\frac{2\left(n+\frac{1}{2}\right)}{1+\left(n+\frac{1}{2}\right)^4-\left(n+\frac{1}{2}\right)^2}\right\}$$

(Dividing numerator and denominator by 16)

$$=\lim_{r\to\infty}\sum_{n=1}^{r}\tan^{-1}\left\{\frac{\left\{\left(n+\frac{1}{2}\right)^2+\left(n+\frac{1}{2}\right)\right\}-\left\{\left(n+\frac{1}{2}\right)^2-\left(n+\frac{1}{2}\right)\right\}}{1+\left\{\left(n+\frac{1}{2}\right)^2-\left(n+\frac{1}{2}\right)\right\}\left\{\left(n+\frac{1}{2}\right)^2+\left(n+\frac{1}{2}\right)\right\}}\right\}$$

$\{\text{Using}: n^4-n^2=\left(n^2-n\right)\left(n^2+n\right)\}$

$$=\lim_{r\to\infty}\sum_{n=1}^{r}\left[\tan^{-1}\left\{\left(n+\frac{1}{2}\right)^2+\left(n+\frac{1}{2}\right)\right\}-\tan^{-1}\left\{\left(n+\frac{1}{2}\right)^2-\left(n+\frac{1}{2}\right)\right\}\right]$$

$\left\{\text{Using}: \tan^{-1}\left(\frac{x-y}{1+xy}\right)=\tan^{-1}x-\tan^{-1}y\ \forall x, y>0\right\}$

$$=\lim_{r\to\infty}\sum_{n=1}^{r}\left[\left(\tan^{-1}\left(\frac{9}{4}+\frac{3}{2}\right)-\tan^{-1}\left(\frac{9}{4}-\frac{3}{2}\right)\right\}+\left\{\tan^{-1}\left(\frac{25}{4}+\frac{5}{2}\right)-\tan^{-1}\left(\frac{25}{4}-\frac{5}{2}\right)\right\}\right.$$

$$+\left\{\tan^{-1}\left(\frac{49}{4}+\frac{7}{2}\right)-\tan^{-1}\left(\frac{49}{4}-\frac{7}{2}\right)\right\}+\ldots+$$

$$\left.\left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2+\left(r+\frac{1}{2}\right)\right\}-\tan^{-1}\left\{\left(r+\frac{1}{2}\right)^2-\left(r+\frac{1}{2}\right)\right\}\right]$$

$$=\lim_{r\to\infty}\left[\left\{\tan^{-1}\left(\frac{15}{4}\right)-\tan^{-1}\left(\frac{3}{4}\right)\right\}+\left\{\tan^{-1}\left(\frac{35}{4}\right)-\tan^{-1}\left(\frac{15}{4}\right)\right\}+\left\{\tan^{-1}\left(\frac{63}{4}\right)-\tan^{-1}\left(\frac{35}{4}\right)\right\}\right.$$

$$+\ldots+\left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2-\tan^{-1}\left(r+\frac{1}{2}\right)\right\}+\left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2-\left(r+\frac{1}{2}\right)\right\}\left.\right]$$

$$=\lim_{r\to\infty}\left[\left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2+\left(r+\frac{1}{2}\right)\right\}-\tan^{-1}\left(\frac{3}{4}\right)\right]$$

$$=\frac{\pi}{2}-\tan^{-1}\left(\frac{3}{4}\right)=\cot^{-1}\left(\frac{3}{4}\right)$$

$$\therefore\cot\alpha=\frac{3}{4}\therefore\operatorname{cosec}\alpha=\sqrt{1+\cot^2\alpha}=\frac{5}{4}$$

$$\therefore\sin\alpha=\frac{4}{5}=\frac{a}{b}$$

$$\therefore\left(a^2+b^2\right)_{\text{least}}=4^2+5^2=16+25=41$$

48. **(97.00)**

$$N=\left(\sqrt[8]{5}+\sqrt[6]{2}\right)^{100}$$

$$T_{r+1}={}^{100}C_r\left(\sqrt[8]{5}\right)^{100-r}\left(\sqrt[6]{2}\right)^r={}^{100}C_r\,(5)^{\frac{100-r}{8}}\frac{r}{2^6}$$

For it to be rational r =12, 36, 60, 84

Hence, 4 rational terms & 97 irrational terms

49. **(192.00)**

12 + 36 + 60 + 84 = 192

50. **(105.00)**

No. of ways to plan. Round 1 = $\frac{8!}{4!(2!)^4}=105$

Prob. of P_4 in final = $\frac{4\times 3!2\times 4!}{\frac{(2!)^3}{105\times 3}}=\frac{4}{35}$

51. **(39.00)**

No. of ways to plan. Round 1 = $\frac{8!}{4!(2!)^4}=105$

Prob. of P_4 in final = $\frac{4\times 3!2\times 4!}{\frac{(2!)^3}{105\times 3}}=\frac{4}{35}$



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if